

README

Super-Resolution Spectral Analysis (SRSA) – MATLAB Implementation

This repository provides a MATLAB implementation of **SRSA (Super-Resolution Spectral Analysis)** together with three demonstration scripts:

1. **High-frequency resolution** – illustrates the ability of SRSA to resolve closely spaced spectral components under noise-free conditions.
2. **Robustness of K** – demonstrates the insensitivity of SRSA to the initial model-order choice (K_{used}) and compares the results with the Fourier spectrum.
3. **Real SG data analysis** – This script demonstrates the analysis of real data.

More detailed comments are provided in the functions **DoSRSA.m** and scripts **Demo1**, **Demo2.m**, **Demo3.m**.

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Repository Structure

```
|— run_demo.m # One-click entry script to reproduce figures
|— README.md # This file
|— README.pdf # This file
|
|— src_SRSA/ # Core SRSA algorithms
|   |— DoSRSA.m
|
|— examples/ # Demonstration scripts and data
|   |— Demo1.m
|   |— Demo2.m
|   |— Demo3.m
```

|
|— figures/ # Output figures generated by the demos

Core function

- `DoSRSA.m`

Core SRSA routine for computing the 2D super-resolution spectrum (interference-response matrix) by scanning window length (L) and candidate frequencies.

Demonstration scripts

- **Demo 1 – High-frequency resolution (noise-free)**

Shows that SRSA can resolve closely spaced frequencies beyond the classical Fourier resolution limit.

- **Demo 2 – Robustness to K**

Shows that SRSA remains stable for model-order guesses and compares results with FFT.

- **Demo 3 – Demonstration of real data analysis**

Using the m1 record from the 2011 event as an example, we reproduce the SR spectral analysis procedure of the OS2 mode.

Quick Start

Requirements

- MATLAB R2024b (tested)
- To facilitate user testing, no additional toolboxes required (only standard functions such as `svd`, `hann`, `hankel`)

Setup

Make sure `DoSRSA.m` is on the MATLAB path.

Run the demos

Simply execute the corresponding scripts in MATLAB:

```
% High-frequency resolution test
run Demo1.m
```

```
% Robustness test
run Demo2.m
```

```
% Demonstration of real data analysis  
run Demo3.m
```

Alternatively, simply run `run_demo.m`

```
% Run all examples  
run run_demo.m
```